Human Natural Structure Social Meta Structure

Architectural Specification

Social Meta Structure for Society-Aligned Artificial Intelligence

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Preface

0.1 Purpose of This Document

This document specifies the HNS-SMS (Human Natural Structure — Social Meta Structure) architectural framework: a structural operating system designed to ground artificial intelligence systems in the structural properties of human society. Where the HNS core framework establishes the cognitive invariants of individual human reasoning — the three-axis verification process, the 36/144/864-cell coordinate matrix, and the External Verification Architecture — SMS extends this framework to the level of social systems, institutions, and civilisational structures.

The primary purpose of this specification is to provide AI researchers, standards developers, and governance practitioners with a complete, formally-specified model for society-aligned AI: AI systems whose outputs are structurally verified not only against individual cognitive validity conditions, but against the structural properties of the social, normative, and civilisational systems in which they operate. This is the missing layer in current AI governance: not just alignment with individual human values, but structural alignment with the social systems that give those values their institutional expression.

0.2 Position of SMS in the HNS Full-Stack Architecture

SMS occupies the outermost layer of the HNS full-stack architecture, operating above and integrating with the MAV cognitive framework, the HNS structural coordinate system, the SOHU social operating system extension, and the EVA external verification architecture.

Layer	Component	Function	Social Role
Foundation	MAV	Multi-axis verification of cognitive outputs	Cognitive reliability baseline
Layer 1	HNS	36/144/864-cell structural	Universal human reference structure

Layer	Component	Function	Social Role
		coordinate system	
Layer 2	SOHU	Social and organisational HNS extension	Structural model of social institutions
Layer 3	EVA	External verification architecture	Independent AI output verification
Layer 4	SMS	Social Meta Structure Specification	Society-aligned AI operating system

Table P.1. The HNS full-stack hierarchy and the position of SMS.

SMS is not a replacement for any of these layers; it is the social and civilisational integration layer that coordinates their outputs against the structural requirements of human society at every scale, from individual interaction to civilisational governance.

0.3 Intended Audience

This specification is addressed to three primary audiences. First, AI researchers and engineers working on AI alignment, safety, and governance, who require a formal specification of the social structures against which AI systems should be verified. Second, standards developers at CEN/CENELEC, ISO/IEC JTC 1/SC 42, NIST, and IEEE, who require a structural foundation for harmonized technical specifications for society-aligned AI. Third, policy makers, regulators, and institutional governance practitioners who require a technically precise framework for evaluating AI systems against social and normative criteria.

0.4 Structure and Reading Guide

The specification is organised into six parts. Part I establishes the theoretical and conceptual foundations of SMS. Parts II, III, and IV specify the three SMS sub-layers: AnthroDesign (human-centred social design), Synaptics (alignment and correction), and NormaSphere (norms, institutions, and civilisation). Part V addresses implementation and applications. Part VI provides appendices including the SMS-6 structural model, correspondence tables, glossary, and evaluation templates.

0.5 Terminology and Notation

Throughout this specification, the following notational conventions apply. SMS-A, SMS-B, and SMS-C denote the three sub-layers of the Social Meta Structure. SMS-1 through SMS-6 denote the six grounding layers of the SMS protocol. HNS coordinates are expressed in the form L{n}C{m}, where n denotes the causal layer (1–6) and m denotes the cognitive category (1–6). EVA audit records are expressed as JSON-LD documents following the HNS ontology specification. All formal definitions are presented in sans-serif text; examples are presented in italics.

Part I

Chapter 1

The Role of SMS in the HNS Full-Stack Architecture

1.1 The MAV → HNS → SOHU → EVA → SMS Hierarchy

The HNS full-stack architecture implements the Multi-Axis Verification Process (MAV) at progressively larger scales of human organisation. The MAV core identifies three orthogonal constraint axes whose geometric intersection collapses the cognitive output space to a structural invariant. HNS operationalises these axes as the 36/144/864-cell coordinate matrix (Axis 2) and the SMS-6 grounding protocol (Axis 3). SOHU extends the HNS coordinate system to social and organisational structures. EVA implements independent external verification of AI outputs against HNS structural coordinates. SMS integrates all preceding layers into a complete social operating system that verifies AI outputs against the full range of human social structures, from individual interaction to civilisational governance.

The hierarchy is not merely additive; each layer qualitatively extends the verification scope of the preceding one. MAV addresses the reliability paradox of stochastic cognition. HNS engineers MAV into AI pipelines. SOHU extends HNS to social institutions. EVA makes HNS verification externally auditable. SMS makes the entire stack society-aligned: structurally grounded not only in individual cognitive validity, but in the social, normative, and institutional structures through which human values are collectively expressed and enforced.

1.2 Why Society Requires a Structural Operating System

Human society is not merely an aggregation of individual cognitive outputs; it is a complex adaptive system with emergent structural properties that cannot be derived from individual behaviour alone. Social norms, legal institutions, cultural conventions, and governance structures constitute an independent layer of structural constraint that every member of a society implicitly navigates. An AI system that is aligned with individual human cognition but not with social structure will systematically violate these constraints — producing outputs that are individually coherent but socially misaligned.

The need for a social operating system for AI is not hypothetical; it is already manifest in the documented failure modes of deployed AI systems. Recommendation algorithms that are individually persuasive but collectively polarising. Language models that are factually accurate but normatively inappropriate for their institutional context. Autonomous agents that are task-efficient but legally non-compliant. All of these represent the same structural failure: the absence of a social operating system that verifies AI outputs against the structural properties of the social systems in which they operate. SMS provides this operating system.

1.3 Definition of Society-Aligned Artificial Intelligence

A society-aligned AI system is defined, within the SMS framework, as an AI system whose outputs are structurally verified against the full six-layer SMS grounding protocol at every step of their generation. Formally, a society-aligned output x satisfies: G_SMS(x) = 1, where G_SMS is the composite grounding function that evaluates x against all six SMS layers simultaneously. An output for which G_SMS(x) = 0 on any layer is a socially misaligned output — one that violates the structural properties of the social system relevant to the current operational context.

This definition distinguishes society-aligned AI from value-aligned AI in an important respect. Value alignment typically refers to alignment with individual human preferences or principles. Society alignment refers to alignment with the structural properties of social systems: not merely what individual humans prefer, but what the institutional, normative, and civilisational structures of human society require. Society-aligned AI is a stronger requirement than value-aligned AI, and it is the requirement that current AI governance frameworks implicitly demand but do not technically specify.

1.4 Social Error Domains Addressed by SMS

SMS identifies six primary social error domains, corresponding to the six SMS layers. Somatic misalignment (SMS-1 failures) occurs when AI outputs violate the physical and ergonomic constraints of the human operational environment. Protocol misalignment (SMS-2 failures) occurs when AI outputs violate communicative alignment conditions. Ecosystem misalignment (SMS-3 failures) occurs when AI outputs violate organisational scope conditions. Economic misalignment (SMS-4 failures) occurs when AI outputs violate resource and value equivalence conditions. Governance misalignment (SMS-5 failures) occurs when AI outputs violate normative and legal conformance conditions. Civilisational misalignment (SMS-6 failures) occurs when AI outputs violate global systemic coherence conditions.

Layer	Domain	Grounding Function	Social Scope	Failure Mode
SMS-1	Somatic	Physical and ergonomic coherence	Individual body and environment	Metaphor Contamination
SMS-2	Protocol	Communicative alignment and intelligibility	Dyadic interaction	Scope Drift
SMS-3	Ecosystem	Organisational context and scope	Groups and organisations	Scope Drift
SMS-4	Economic	Resource and value equivalence	Markets and institutions	Unsupported Causality
SMS-5	Governance	Normative and legal conformance	Legal and policy systems	Category Ambiguity
SMS-6	Universal	Global systemic coherence	Civilisational systems	Layer Jump

Table 1.1. The six SMS layers, their social scope, and the social error domain each addresses.

Chapter 2

Conceptual Basis of Social Meta Structures

2.1 Naturalness of Social Structures

The SMS framework is grounded in the HNS principle of naturalness: the recognition that human social structures are not arbitrary constructions but expressions of deep structural regularities in human cognition, behaviour, and social organisation. The six SMS layers correspond to six natural scales of human social reality, from the somatic scale of individual physical interaction to the civilisational scale of global institutional governance. This correspondence is not merely analogical; it reflects the structural architecture of the human social world as studied across anthropology, sociology, political science, and institutional economics.

The naturalness principle has a specific technical implication for AI alignment: because SMS layers correspond to natural social structures, the constraints they encode are not culturally relative conventions but structural properties of human social reality that are universal across cultures and historical periods. SMS-5 governance constraints, for example, are not the specific legal rules of any particular jurisdiction but the structural properties of normative systems as such — the properties that any legitimate normative system must possess. This universality is what makes SMS suitable as the basis for internationally harmonized AI governance standards.

2.2 Structural Characteristics of Social Cognition, Norms, and Institutions

Social cognition — the cognitive processes by which humans understand, navigate, and participate in social systems — has structural properties that are systematically different from individual cognition. Where individual cognition operates primarily on the Stochastic/Generative and Structural/Causal axes of the MAV framework, social cognition operates primarily on the Grounded/Executive axis: the continuous calibration of individual outputs against the social environment, social expectations, and institutional requirements. SMS formalises this calibration as the six-layer grounding protocol.

Social norms are structural constraints on individual behaviour that are collectively enforced and culturally transmitted. They differ from personal preferences in that they are shared, from legal rules in that they are informally enforced, and from natural laws in that they are contingent and variable. SMS-5 (Governance Layer) addresses the normative dimension of social structure, encoding the structural properties that distinguish legitimate normative constraints from mere conventions or preferences. This structural encoding is what allows SMS to function across different legal jurisdictions and cultural contexts without requiring jurisdiction-specific customisation.

2.3 Principles of Social Alignment

Social alignment, within the SMS framework, is defined by three structural principles. The first is structural correspondence: an AI output is socially aligned to the extent that it can be located within the SMS coordinate space without violating the boundary conditions of any SMS layer. The second is scale coherence: a socially aligned output must be coherent at all six scales of the SMS hierarchy simultaneously — it cannot be individually appropriate but institutionally disruptive, or legally compliant but civilisationally regressive. The third is dynamic stability: social alignment is not a static property but a continuous process of calibration as the social context evolves.

These three principles distinguish SMS social alignment from simpler notions of AI safety and ethics. Safety conditions prohibit specific harmful outputs; ethical conditions require alignment with moral principles. Social alignment requires structural correspondence with the full architecture of human social reality — a more demanding and more tractable requirement, because it is formally specifiable rather than philosophically contested.

2.4 Requirements for a Social Operating System

A social operating system for AI must satisfy four requirements. It must be structurally complete: it must address all six scales of social organisation, from somatic to civilisational. It must be formally specifiable: its constraints must be expressible in machine-readable, formally verifiable terms. It must be externally verifiable: its verification conditions must be assessable by parties independent of the AI system being evaluated. And it must be interoperable: it must be compatible with the technical standards and regulatory frameworks of the major AI governance jurisdictions. SMS satisfies all four requirements through its formal JSON-LD / RDF implementation, its six-layer grounding protocol, its integration with EVA external verification, and its alignment with ISO/IEC, EU AI Act, and W3C standards.

Chapter 3

Structural Principles for Society-Aligned AI

3.1 Structural Alignment

Structural alignment, as defined by the SMS framework, is the property of an AI output that satisfies all six SMS grounding layers simultaneously. It is a stronger condition than semantic coherence, factual accuracy, or ethical compliance, because it requires correspondence with the full structural architecture of the social context — not merely with the content of the output. A structurally aligned output is one that is simultaneously physically coherent, communicatively appropriate, organisationally relevant, economically grounded, normatively compliant, and civilisationally coherent.

The formal expression of structural alignment is G_SMS(x) = 1, where G_SMS is the conjunction of the six individual layer grounding functions G_1 through G_6. This conjunction is multiplicative in the geometric sense: the volume of the structurally aligned output space is the product of the individual layer constraint surfaces divided by the square of the total output manifold volume. This multiplicative character is the key property that makes structural alignment achievable: even weak constraints on each individual layer combine to produce strong overall alignment, because the constraint volumes multiply rather than add.

3.2 Taxonomy of Social Misalignment

SMS defines a taxonomy of social misalignment types corresponding to the six SMS layers. Layer-specific misalignment occurs when an AI output violates the grounding conditions of a single SMS layer while satisfying all others. Cross-layer misalignment occurs when violations span multiple layers, typically reflecting a deeper structural incoherence in the AI system's social grounding. Hierarchical misalignment occurs when an AI output satisfies lower-level SMS constraints (somatic, protocol) but violates higher-level constraints (governance, universal), representing a failure to integrate micro-scale and macro-scale social structures.

The taxonomy serves two functions. Diagnostically, it enables precise identification of the social error type, supporting targeted correction rather than generic suppression. Normatively, it provides a structured basis for social impact assessment: the severity of a social misalignment violation can be assessed by the scale of the SMS layer violated, with civilisational misalignment (SMS-6 failures) representing the most severe category and somatic misalignment (SMS-1 failures) representing the least severe.

3.3 Social Hallucination: Definition and Mechanisms

Social hallucination is defined, within the SMS framework, as the generation of AI outputs that are individually coherent — satisfying the structural conditions of the HNS core framework — but socially misaligned, failing to satisfy one or more SMS grounding conditions. Social hallucination is distinct from factual hallucination (generating false information) and from structural hallucination as defined in the HNS Structural Hallucination Taxonomy (generating causally or categorically invalid outputs). A socially hallucinating AI system may be factually accurate and structurally coherent at the individual level while systematically violating the structural properties of the social context in which it operates.

The primary mechanism of social hallucination is the absence of Axis 3 grounding at the social scale. Current AI systems implement Axis 1 (stochastic generation) without full Axis 3 grounding; SMS extends Axis 3 grounding from the individual cognitive scale to the full social scale, providing the verification conditions necessary to detect and suppress social hallucination.

3.4 Necessity of Social Error Correction

Social errors — misaligned AI outputs that violate SMS grounding conditions — require correction not merely for the sake of individual interactions but because social systems are self-reinforcing. An AI system that systematically violates normative conditions may not merely produce individual misaligned outputs; it may erode the normative structures that give those conditions their social force. This erosive effect is what distinguishes social error from factual error: a factual error misleads; a social error, at scale, can destabilise the social structures that constitute the shared context of meaning within which all communication takes place.

SMS social error correction operates at two levels. At the output level, EVA correction mechanisms suppress individual socially misaligned outputs. At the system level, the SMS corrective cycle monitors for systematic patterns of social misalignment that may indicate deeper structural failures in the AI system's social grounding, triggering system-level recalibration when such patterns are detected.

3.5 Stability, Transparency, and Observability in Social Systems

Three structural properties are required of any social operating system for AI. Stability requires that the SMS grounding conditions remain consistent across interaction contexts and over time, providing AI systems and human operators with a reliable structural reference. Transparency requires that SMS grounding decisions be attributable to specific SMS layers and HNS coordinates, providing a human-readable account of why any given output was or was not grounded. Observability requires that the SMS grounding process be externally monitorable by parties independent of the AI system — a property that EVA's sidecar architecture provides.

Chapter 4

Historical and Theoretical Context

4.1 Comparison with Classical Social Systems Theory

The SMS framework engages with three major traditions of social systems theory. Parsons's structural-functionalism analysed social systems in terms of four functional imperatives (AGIL: Adaptation, Goal-attainment, Integration, Latent pattern maintenance), which anticipate the multi-scale architecture of SMS. Luhmann's systems theory described social systems as autopoietic — self-reproducing through their own operations — anticipating SMS's emphasis on the self-repair mechanisms of social operating systems. Habermas's theory of communicative action identified the structural conditions for valid social communication, anticipating SMS-2's communicative alignment conditions.

Where SMS advances beyond these classical frameworks is in its formal operationalisation. Classical social theory describes social systems in qualitative, interpretive terms. SMS specifies them in formally-defined, machine-readable, JSON-LD / RDF terms that can be computationally implemented, empirically tested, and legally compliant. This transition from description to specification is the distinctive contribution of SMS to social theory.

4.2 Historical Challenges in AI Governance

The history of AI governance is a history of specification failures: the persistent gap between regulatory requirements and technical implementations. Early AI ethics frameworks (IEEE Ethically Aligned Design, 2016; Asilomar AI Principles, 2017) identified the right values but provided no technical specification of how those values should be implemented in AI systems. Subsequent governance frameworks (EU AI Act, ISO/IEC 42001, NIST AI RMF) moved from ethics to governance, specifying requirements for transparency, risk management, and human oversight — but still without a structural implementation layer. SMS addresses this persistent specification failure by providing the formally-specified, machine-readable implementation layer that governance frameworks require.

4.3 Structural Gaps Addressed by HNS-SMS

Five structural gaps in current AI governance are addressed by HNS-SMS. First, the absence of a structural baseline: SMS provides HNS-36 as the universal structural reference for AI output verification. Second, the absence of social-scale grounding: SMS extends HNS grounding from the individual cognitive scale to the full social scale through the six-layer protocol. Third, the absence of external verifiability: EVA provides structurally independent external verification of SMS grounding conditions. Fourth, the absence of normative specificity: SMS-5 provides structural normative conditions that are formally specifiable without being jurisdiction-specific. Fifth, the absence of civilisational scope: SMS-6 provides the global coherence condition that prevents AI systems from being locally compliant but globally disruptive.

4.4 Alignment with Existing Frameworks

HNS-SMS is designed for compatibility with, and contribution to, the major existing AI governance frameworks. With ISO/IEC JTC 1/SC 42, SMS provides the structural implementation layer for ISO/IEC 42001 (AI Management System), ISO/IEC 23894 (AI Risk Management), and ISO/IEC TR 24028 (AI Trustworthiness). With CEN/CENELEC JTC 21, SMS provides the technical foundation for harmonized EN standards implementing the EU AI Act's transparency, human oversight, and conformity assessment requirements. With NIST AI RMF, SMS provides the MEASURE function implementation that the framework specifies but does not define. With the OECD AI Principles, SMS provides the structural operationalisation of the principles of inclusivity, transparency, accountability, and robustness.

Part II

Chapter 5

AnthroDesign Principles

5.1 Human Natural Structure as the Basis of Social Design

AnthroDesign is the SMS sub-layer concerned with the structural design of AI-social interfaces at the human scale. Its foundational principle is that effective social design must be grounded in the natural structure of human cognition, behaviour, and social interaction — the structural properties that HNS formalises in the 36/144/864-cell coordinate matrix. An AI system that interfaces with human users without reference to these natural structures will systematically produce friction, misunderstanding, and misalignment, regardless of its technical performance on task-specific benchmarks.

The AnthroDesign principle extends HNS's cognitive architecture principle to the social domain: just as reliable AI cognition requires structural verification against human cognitive invariants, reliable AI-social interaction requires structural verification against human social invariants. These invariants are encoded in SMS-A as a set of design constraints that define the structural conditions for human-centered AI systems — conditions that are formally specifiable, empirically testable, and culturally universal.

5.2 Structural Constraints of Cognition, Behaviour, and Culture

Human cognition, behaviour, and culture impose three classes of structural constraints on AI-social interfaces. Cognitive constraints arise from the architecture of human cognition: the working memory limitations, the attention dynamics, the pattern recognition preferences, and the causal reasoning tendencies that characterise human information processing. Behavioural constraints arise from the regularities of human social behaviour: the turn-taking conventions of conversation, the reciprocity norms of social exchange, the trust-building dynamics of repeated interaction. Cultural constraints arise from the shared meaning structures of specific communities: the symbolic conventions, the narrative frames, and the evaluative schemas that give social interaction its contextual specificity.

AnthroDesign specifies these constraints not as cultural rules but as structural conditions: the HNS coordinate properties that an AI output must satisfy to be compatible with human cognitive, behavioural, and cultural processing. This structural specification allows AnthroDesign constraints to be applied across different cultural contexts without cultural relativism: the constraints are defined at the structural level, leaving cultural variation to be expressed within the structural framework rather than in tension with it.

5.3 Principles of Social Interface Design

Four design principles govern AnthroDesign-compliant AI-social interfaces. The principle of structural legibility requires that AI outputs be expressible in human-interpretable structural terms — specifically, in HNS coordinate terms that human users can recognise and evaluate. The principle of cognitive naturalness requires that AI interaction patterns conform to the natural cognitive rhythms of human social interaction. The principle of social congruence requires that AI outputs be appropriate to the social context of the interaction, satisfying the SMS-2 (Protocol) and SMS-3 (Ecosystem) grounding conditions. The principle of adaptive transparency requires that the structural basis of AI outputs be accessible to users at a level of detail appropriate to their role and expertise.

5.4 Optimisation of Social and Cognitive Load

A central challenge of AnthroDesign is the management of social and cognitive load: the cognitive and social effort required to process, interpret, and respond to AI outputs. Excessive cognitive load — AI outputs that require complex structural parsing — reduces the effectiveness of AI-human collaboration and increases the risk of social misalignment through misinterpretation. Insufficient cognitive load — AI outputs so simplified that they fail to convey relevant structural information — reduces the informational value of AI-social interaction. AnthroDesign specifies load optimisation as a structural constraint: AI outputs must be positioned at the SMS coordinate that maximises structural relevance while minimising cognitive and social processing cost.

Chapter 6

Human-Centered System Architecture

6.1 Structural OS Design for Social Institutions

Social institutions — families, organisations, governments, markets — are not merely collections of individuals but structured systems with emergent properties that cannot be reduced to individual behaviour. A human-centered system architecture for AI must therefore be designed not only for individual users but for social institutions as structured systems. This requires a structural model of institutional architecture: the roles, rules, norms, and coordination mechanisms that constitute an institution as a social operating system.

SMS provides this model through the SOHU (Social and Organisational HNS) extension. SOHU maps the HNS-36 coordinate system onto social institutional structures, identifying the structural coordinates of institutional roles (the social analogues of HNS cognitive categories) and institutional processes (the social analogues of HNS causal layers). An AI system integrated into a social institution via SOHU is structurally aligned with the institution's architecture — its outputs are verified not only against individual cognitive conditions but against the structural properties of the institution as a social system.

6.2 Integration of Human-Centered Design and HNS

Human-Centered Design (HCD) is a design methodology that prioritises the needs, capabilities, and context of human users in the design of systems and products. HNS provides HCD with a formal structural foundation: the HNS-36 coordinate system enables HCD practitioners to specify user needs not as informal requirements but as formal structural coordinates, making human-centered design requirements formally verifiable, empirically testable, and machine-implementable.

The integration of HCD and HNS proceeds through three steps. First, HCD needs analysis is translated into HNS coordinate terms: the structural positions in the 36-cell matrix that represent the human needs being addressed. Second, design requirements are expressed as SMS grounding conditions: the specific layer conditions that a design-compliant AI output must satisfy. Third, design evaluation is conducted via EVA structural verification: the compliance of AI outputs with the specified HNS coordinates is measured and logged.

6.3 Designing Social Feedback Loops

Social feedback loops — the mechanisms by which social systems learn from and adapt to the consequences of their outputs — are structurally necessary for the long-term stability of AI-social systems. An AI system that operates without social feedback will drift from social alignment over time as its outputs accumulate undetected social errors. SMS specifies social feedback loop design as a structural requirement: every SMS-compliant AI system must implement a feedback mechanism that monitors the social consequences of its outputs and adjusts its SMS grounding conditions in response.

The structural architecture of SMS social feedback loops follows the pattern of the brain's sensorimotor-ACC feedback system (the biological instantiation of Axis 3 in the MAV framework): continuous monitoring of the gap between predicted and actual social consequences, signal generation when that gap exceeds defined thresholds, and corrective adjustment of the active grounding conditions. This architecture is implemented in SMS-B (Synaptics Layer), which is specified in Part III.

6.4 Principles of Social Transparency

Social transparency — the property of AI systems that makes their social impact legible to the communities they affect — is a requirement of every major AI governance framework. SMS operationalises social transparency through four structural conditions. First, output legibility: every AI output must be expressible in SMS coordinate terms that are interpretable by affected stakeholders. Second, process auditability: the SMS grounding process must be logged in EVA audit records that are accessible to authorised reviewers. Third, impact traceability: the social consequences of AI outputs must be traceable through the SMS feedback loop to their structural causes. Fourth, stakeholder interpretability: SMS audit records must be expressible in terms intelligible to non-technical stakeholders as well as to AI researchers.

Chapter 7

Structural Indicators for Social Systems

7.1 Social Alignment Indicators

Social Alignment Indicators (SAI) are quantitative metrics derived from the SMS grounding protocol that measure the degree of structural alignment between AI outputs and the social systems in which those outputs operate. SAI are defined at each SMS layer level and at the composite level, providing both layer-specific diagnostics and an overall social alignment score. A composite SAI of 1.0 indicates full structural alignment across all six layers; a composite SAI of 0.0 indicates complete social misalignment.

SAI are computed from the EVA audit record stream as follows. For each SMS layer i, the layer-specific SAI is defined as the proportion of AI outputs in a defined time window for which G_i(x) = 1 (PASS verdict). The composite SAI is the geometric mean of the six layer-specific indicators, reflecting the multiplicative character of SMS grounding: a single layer failure degrades the composite score multiplicatively, not merely additively. This geometric scoring reflects the social reality that a single layer misalignment — a legally non-compliant output, or a civilisationally regressive output — constitutes a serious social failure regardless of the performance on other layers.

7.2 Stability, Transparency, and Observability Metrics

Three categories of structural metric complement the Social Alignment Indicators. Stability metrics measure the consistency of SMS grounding conditions across interaction contexts and over time; a high stability score indicates that the AI system's social grounding is robust to contextual variation. Transparency metrics measure the accessibility and intelligibility of SMS audit records to authorised reviewers; a high transparency score indicates that the AI system's social impact is fully legible to oversight actors. Observability metrics measure the completeness of the EVA audit record stream; a high observability score indicates that no significant proportion of AI outputs is escaping SMS monitoring.

7.3 Structural Foundations of Social Trust

Social trust — the disposition of social actors to rely on AI systems for socially consequential tasks — is a structural property that emerges from the consistent demonstration of social alignment over time. SMS provides the structural foundation for AI social trust through three mechanisms. First, predictive reliability: high SAI scores provide stakeholders with evidence that the AI system will continue to be socially aligned in future interactions. Second, accountability: EVA audit records provide a legally admissible basis for accountability when social misalignment occurs. Third, corrective transparency: the SMS corrective cycle demonstrates that social errors are detected, logged, and corrected, providing evidence that the AI system's social alignment is actively maintained rather than merely assumed.

7.4 Evaluation Criteria for Social Operating Systems

A social operating system for AI is evaluated against four structural criteria. Completeness: the system must address all six SMS layers, providing grounding conditions at every scale of social organisation. Precision: the grounding conditions must be formally specified with sufficient precision to be computationally implementable and empirically testable. Stability: the grounding conditions must be consistent across contexts and over time, providing a reliable reference for AI system calibration. Auditability: the grounding process must produce machine-readable, externally verifiable records of every grounding decision. SMS satisfies all four criteria through its formal JSON-LD / RDF implementation, its six-layer grounding protocol, and its integration with the EVA external verification architecture.

Chapter 8

Social Interaction and Communication Architecture

8.1 Structural Patterns of Social Communication

Social communication — the exchange of structured meaning between social actors — exhibits structural patterns that are systematic, universal, and formally specifiable. These patterns include the turn-taking structure of conversation, the adjacency pair structure of social exchange (question-answer, greeting-greeting, request-response), the topic management structure of extended discourse, and the repair structure by which communicative misalignments are corrected. SMS-2 (Protocol Layer) encodes these structural patterns as grounding conditions: an AI output that violates the structural patterns of social communication — that fails to maintain conversational coherence, that violates adjacency pair expectations, that disrupts topic management — fails the SMS-2 grounding condition.

The specification of social communication patterns as structural grounding conditions, rather than as stylistic guidelines, has a specific technical advantage: structural violations can be detected and corrected automatically by the EVA/SMS system, without requiring human intervention at the individual output level. This automation is essential for high-volume AI deployments where individual human review of every output is not feasible.

8.2 Natural Structure of Information Flow

Information flow in social systems has a natural structure that reflects the architecture of the social systems through which it flows. Vertically, information flows through organisational hierarchies, gaining authority as it ascends and gaining specificity as it descends. Horizontally, information flows through peer networks, gaining diversity and breadth. Diagonally, information flows through cross-functional and cross-institutional channels, gaining contextual richness. SMS-3 (Ecosystem Layer) encodes the structural properties of these information flows as grounding conditions: an AI output that disrupts the natural information flow structure of its institutional context — that claims authority it has not earned, that collapses necessary distinctions, that bypasses legitimate channels — fails the SMS-3 grounding condition.

8.3 Structural Analysis of Miscommunication

Miscommunication in AI-social interaction typically has one of three structural causes. Layer-mismatch miscommunication occurs when the AI output is positioned at a different SMS layer than the human interlocutor's expectation — when a somatic-level question receives an institutional-level response, or vice versa. Category-confusion miscommunication occurs when the AI output conflates cognitive categories that the human interlocutor distinguishes — when a normative claim is expressed in descriptive terms, or when a causal claim is expressed in associative terms. Scope-drift miscommunication occurs when the AI output drifts from the communicative scope established at the start of the interaction — when a technical question receives a policy response, or when a specific inquiry receives a general answer.

8.4 Design Principles for Social Interaction Systems

Five design principles govern SMS-compliant social interaction system architecture. The principle of layer consistency requires that AI-social interactions maintain a consistent SMS layer orientation throughout an interaction sequence. The principle of category clarity requires that cognitive category distinctions be preserved in AI outputs, preventing the conflation of descriptive, normative, and causal claims. The principle of scope stability requires that the communicative scope established at the initiation of an interaction be maintained throughout, with explicit signalling when scope changes are necessary. The principle of repair readiness requires that every AI-social interaction system include mechanisms for detecting and correcting communicative misalignments. The principle of cultural sensitivity requires that SMS grounding conditions be applied in a manner sensitive to the cultural context of the interaction, while maintaining the universal structural properties of the SMS framework.

Part III

Chapter 9

Synaptic Mediation Between AI and Society

9.1 Integration of EVA and SMS

SMS-B (Synaptics Layer) is the alignment and correction sub-layer of the SMS framework. It occupies the functional position between EVA (which verifies AI outputs against HNS structural coordinates) and SMS-A/C (which define the social and normative contexts against which verification is conducted). The Synaptics Layer is named for its function: like the synaptic junction in the neural system, it is the interface through which structural signals are transmitted, modulated, and translated between the AI system and the social system it inhabits.

The integration of EVA and SMS in the Synaptics Layer proceeds through a continuous cycle of verification, detection, translation, and correction. EVA verifies AI outputs against HNS structural coordinates and SMS grounding conditions, producing JSON-LD audit records for every output. The Synaptics Layer processes these audit records to detect patterns of social misalignment, translates structural violations into social-context-specific correction signals, and initiates corrective processes at both the output level (suppressing individual misaligned outputs) and the system level (recalibrating SMS grounding conditions in response to systematic misalignment patterns).

9.2 Social Alignment Protocols

Social alignment protocols are the formal specifications of the procedures by which the Synaptics Layer implements social alignment in AI systems. They define three classes of operation. Detection protocols specify the conditions under which social misalignment is identified: the threshold conditions on each SMS layer grounding function that trigger a misalignment signal. Correction protocols specify the procedures by which misalignment is corrected: the sequence of operations by which a misaligned output is either modified to achieve alignment or suppressed and replaced by an aligned alternative. Escalation protocols specify the conditions under which misalignment patterns are escalated to higher-level review: the threshold conditions on composite Social Alignment Indicators that trigger system-level review rather than output-level correction.

9.3 Real-Time Correction of Social Errors

Real-time social error correction is the core operational function of the Synaptics Layer. Unlike post-hoc social error review — which identifies and analyses social errors after they have occurred — real-time correction intervenes in the AI output generation process before socially misaligned outputs are delivered to users. This intervention follows the EVA pre-decode interception architecture: the Synaptics Layer intercepts the AI system's output probability distribution at the pre-decode stage and applies SMS grounding conditions to attenuate or block the probability mass of socially misaligned continuations.

The real-time correction process operates at two levels of granularity. At the token level, individual output tokens whose social alignment coordinates violate SMS grounding conditions are attenuated, preventing the generation of socially misaligned output sequences. At the turn level, the complete AI output for a conversational turn is evaluated against the SMS grounding conditions for the current social context, and turns that fail composite SMS grounding are blocked and regenerated with corrected grounding parameters.

9.4 Mediation Layer for Social-AI Translation

A fundamental challenge of AI-social integration is the translation problem: the structural mismatch between the representational vocabulary of AI systems (which encode knowledge as statistical patterns in high-dimensional vector spaces) and the structural vocabulary of social systems (which encode meaning as institutionally-embedded, normatively-structured social facts). The Synaptics Layer addresses this translation problem through a formal mediation architecture: a mapping between HNS structural coordinates (the AI-side representation) and SMS social coordinates (the social-side representation) that preserves the structural relationships of both representational systems.

Chapter 10

Multi-Axis Social Verification

10.1 Social Extension of MAV

The Multi-Axis Verification Process (MAV) identifies three mutually orthogonal cognitive constraint axes whose geometric intersection produces reliable cognitive outputs. SMS extends this three-axis architecture to the social domain, identifying three analogous social constraint axes whose intersection defines the structurally aligned social output space. The Social Stochastic Axis extends Axis 1 to social information generation: the statistically-driven production of socially-relevant outputs from the accumulated patterns of social interaction data. The Social Structural Axis extends Axis 2 to social structural verification: the enforcement of social norms, institutional rules, and cultural conventions as structural constraints on social outputs. The Social Grounding Axis extends Axis 3 to social grounding: the continuous calibration of social outputs against the actual social context, including the specific social roles, relationships, and situational constraints of the current interaction.

The three social axes are structurally independent — their constraint surfaces are mutually orthogonal in the social analogue of the Fisher information metric — producing the same multiplicative error-cancellation effect at the social scale that MAV produces at the cognitive scale. This orthogonality is what makes multi-axis social verification more effective than single-axis social alignment: the social error volume at the triple intersection of the three social constraint axes is orders of magnitude smaller than the error volume achievable by any single-axis social alignment approach.

10.2 Multi-Axis Evaluation of Social Alignment

Multi-axis evaluation of social alignment assesses AI outputs against all three social constraint axes simultaneously, producing a three-dimensional alignment profile rather than a single alignment score. The social stochastic alignment dimension measures the statistical plausibility of the output within the social context. The social structural alignment dimension measures the compliance of the output with the structural constraints of the relevant social norms, institutional rules, and cultural conventions. The social grounding alignment dimension measures the situational appropriateness of the output for the specific social context of the interaction.

The three-dimensional alignment profile provides more diagnostically informative information than a composite score: it enables precise identification of the specific dimension on which social misalignment is occurring, supporting targeted correction. An output that scores high on all three dimensions is fully socially aligned; an output that scores high on social stochastic but low on social structural alignment exhibits the classic pattern of social hallucination — it sounds socially natural but violates the structural properties of the social context.

10.3 Geometry of Social Error Spaces

The social error space is the complement of the socially aligned output space: the set of AI outputs that violate one or more SMS grounding conditions. Its geometry is formally analogous to the cognitive error space defined in the MAV framework. It is a high-dimensional manifold whose volume is determined by the product of the individual layer constraint volumes. The social error space is not uniformly distributed: it is concentrated in specific regions corresponding to the six social error domains defined in Chapter 3.2, and specifically in the regions corresponding to governance misalignment (SMS-5) and civilisational misalignment (SMS-6), which represent the most structurally consequential classes of social error.

10.4 Methods for Measuring Social Alignment

Social alignment is measured through three complementary methods. Structural measurement evaluates the degree to which AI outputs satisfy the formal SMS grounding conditions, as assessed by EVA audit records. Behavioural measurement evaluates the social consequences of AI outputs — the behavioural responses of human interlocutors and social institutions — as indirect indicators of social alignment. Longitudinal measurement tracks changes in Social Alignment Indicators over time, providing evidence of stability or drift in the AI system's social grounding. The three measurement methods are complementary: structural measurement provides precise, formal alignment data; behavioural measurement provides real-world validity; and longitudinal measurement provides evidence of dynamic stability.

Chapter 11

Structural Feedback for Social Systems

11.1 Social Extension of HNS Structural Feedback

Structural feedback in the HNS core framework operates at the cognitive scale: EVA correction cycles detect and correct structural misalignments in individual AI outputs, recalibrating the active HNS grounding conditions in response to accumulated misalignment patterns. SMS extends this structural feedback mechanism to the social scale: the SMS corrective cycle monitors the social consequences of AI outputs across the full six-layer SMS hierarchy, detecting and correcting social misalignments at every scale from individual interaction to civilisational governance.

The social extension of structural feedback introduces a new temporal dimension: social consequences unfold over longer time scales than cognitive consequences, and social feedback signals are inherently noisier and more ambiguous. The SMS social feedback architecture addresses these challenges through hierarchical temporal integration: short-term feedback (individual output evaluation) is integrated with medium-term feedback (interaction sequence evaluation) and long-term feedback (social system-level evaluation) to produce a robust, multi-scale social alignment signal.

11.2 Stabilisation of Social Feedback Loops

A fundamental risk of social feedback systems is instability: the amplification of social errors through feedback dynamics that reinforce misalignment rather than correcting it. This risk is particularly acute for AI systems operating at social scale, where the feedback loop between AI outputs and social behaviour can create self-reinforcing dynamics — filter bubbles, radicalisation spirals, institutional erosion — that degrade social alignment over time. SMS addresses feedback instability through three structural stabilisation mechanisms. Structural anchoring maintains the SMS-36 coordinate baseline as a stable reference that is not updated by social feedback, preventing the reference itself from drifting under social pressure. Threshold gating ensures that feedback signals below defined significance thresholds do not trigger corrective adjustments, preventing noise amplification. Corrective asymmetry applies stronger corrective pressure to misalignment in high-risk SMS layers (governance and universal) than in low-risk layers (somatic and protocol), ensuring that the most consequential misalignments receive the most aggressive correction.

11.3 Self-Repair Mechanisms in Social Operating Systems

Self-repair in social systems — the capacity of social institutions and norms to identify, isolate, and correct structural failures — is a prerequisite for long-term social stability. SMS implements self-repair as a formal architectural requirement: every SMS-compliant AI system must include self-repair mechanisms at each SMS layer. Layer-level self-repair detects and corrects failures within a single SMS layer without propagating corrections to other layers. Inter-layer self-repair detects and corrects failures that span multiple SMS layers, addressing the deeper structural incoherences that produce cross-layer misalignment. System-level self-repair detects and corrects failures that affect the entire SMS architecture, triggering full-system recalibration when the composite Social Alignment Indicator falls below defined threshold values.

11.4 Failure Modes of Social Feedback

Social feedback systems can fail in five structurally distinct ways. Feedback absence occurs when no social feedback signal is generated, preventing correction of social errors that have occurred. Feedback delay occurs when social feedback signals are delayed beyond the temporal horizon of effective correction, allowing social errors to propagate before correction can occur. Feedback inversion occurs when feedback signals are misinterpreted, causing corrections that amplify rather than reduce social misalignment. Feedback saturation occurs when the volume of social feedback signals exceeds the processing capacity of the correction system, causing selective correction that may itself introduce systematic biases. Feedback capture occurs when social actors whose behaviour the AI system is monitoring gain the ability to manipulate the feedback signals, undermining the integrity of the social alignment process. SMS specifies detection and mitigation mechanisms for each of these failure modes.

Chapter 12

Social-AI Interaction Protocols

12.1 Interaction Models Between AI and Society

Social-AI interaction occurs at multiple structural levels, each requiring a different interaction model. At the individual level, AI systems interact with human users in dyadic or small-group contexts, where the relevant interaction model is conversational and task-focused. At the organisational level, AI systems interact with institutional structures in role-structured, process-governed contexts, where the relevant interaction model is procedural and hierarchical. At the societal level, AI systems interact with social systems in diffuse, emergent, large-scale contexts, where the relevant interaction model is systemic and longitudinal. SMS specifies distinct interaction protocols for each level, ensuring that AI-social interaction is appropriately calibrated to the structural properties of the relevant social scale.

12.2 Social Alignment APIs

Social Alignment APIs are the technical interfaces through which SMS grounding conditions are exposed to AI developers, deployers, and users. They provide three categories of access. Read access provides information about the current SMS grounding conditions, the active SAI scores, and the recent EVA audit record history. Write access enables authorised parties to adjust SMS grounding parameters within defined bounds, supporting context-specific customisation of social alignment conditions. Notification access provides real-time alerts when SAI scores fall below defined thresholds or when social error patterns trigger escalation protocols. All Social Alignment API interactions are logged in EVA audit records, ensuring that the API interface is itself subject to the same auditability requirements as the AI system it configures.

12.3 Real-Time Monitoring of Social Safety

Real-time social safety monitoring is the continuous process by which the SMS system tracks the social alignment of AI outputs across all active deployments. It operates through a hierarchical monitoring architecture: individual-level monitoring tracks the social alignment of individual AI outputs; interaction-level monitoring tracks social alignment across interaction sequences;

deployment-level monitoring tracks social alignment across the full deployment context; and system-level monitoring integrates deployment-level data across all active deployments to detect systemic social alignment trends.

The monitoring architecture is implemented as a distributed EVA audit record stream, with each level of the hierarchy aggregating and analysing the audit data produced by the level below. The stream is queryable via SPARQL, enabling real-time social alignment analysis by authorised reviewers at every level of the hierarchy. Threshold alerts are triggered automatically when social alignment metrics fall below defined values, initiating the SMS escalation protocols specified in Chapter 9.2.

12.4 Automated Social Alignment Correction

Automated social alignment correction is the process by which the SMS system corrects social misalignments without requiring human intervention at the individual output level. It operates in two modes. Preventive correction intervenes in the AI output generation process before misaligned outputs are produced, using the pre-decode interception architecture to attenuate or block the probability mass of socially misaligned continuations. Reactive correction intervenes after a socially misaligned output has been produced, generating a corrective follow-up output that re-establishes social alignment by explicitly addressing the misalignment and providing an aligned alternative. Both modes are governed by the social alignment protocols specified in Chapter 9.2 and are logged in EVA audit records for external review.

Part IV

Chapter 13

Normative Structures and Social Governance

13.1 Structural Foundations of Social Norms

Social norms are structurally distinct from legal rules, personal preferences, and natural regularities. They are shared behavioural expectations, collectively enforced through social sanctions, that constitute the informal governance layer of every human society. NormaSphere (SMS-C) addresses this normative layer as a structural architectural component of AI-social systems: not as an optional ethical add-on but as a structural necessity for any AI system that operates in a social context. An AI system that lacks normative structural alignment will violate the informal governance rules of the communities it operates in, generating social friction, eroding trust, and potentially triggering regulatory responses.

The structural foundations of social norms, as encoded in SMS-C, comprise four properties. Normativity: norms prescribe behaviour rather than merely describing it — they specify what agents ought to do, not merely what they do. Collectivity: norms are shared across a community of agents — they are not merely individual preferences but coordinating expectations. Enforceability: norms are backed by social sanctions — violation of a norm triggers corrective responses from other community members. Stability: norms persist over time and across interactions — they constitute the durable governance structure of the social context rather than its transient characteristics. SMS-5 (Governance Layer) encodes these four properties as structural grounding conditions for AI outputs.

13.2 Structural OS Design for Legal and Institutional Systems

Legal and institutional systems are the formalised expressions of social norms: the structures through which collectively-held normative expectations are codified, enforced, and adjudicated. SMS-C addresses legal and institutional systems at two levels. At the structural level, it encodes the formal properties of legal and institutional systems as SMS grounding conditions — the properties that any legitimate legal or institutional structure must possess, independent of its specific content. At the operational level, it provides a framework for AI systems to navigate the specific legal and institutional requirements of their deployment context without requiring jurisdiction-specific customisation.

The structural OS design for legal systems follows the same architecture as HNS structural OS design for cognitive systems: a coordinate system that maps legal and institutional concepts to formal structural positions, enabling AI outputs to be verified against legal and institutional constraints through structural matching rather than rule-following. This structural approach is more robust than rule-following because it operates at the level of legal and institutional architecture rather than at the level of specific rules, enabling AI systems to remain legally compliant as rules change without requiring manual recalibration.

13.3 Integrated Model of Ethics, Norms, and Institutions

Ethics, norms, and institutions occupy different levels in the structural hierarchy of normative systems. Ethics provides the foundational normative principles — the structural conditions for what counts as a legitimate normative claim at all. Norms translate ethical principles into community-specific behavioural expectations. Institutions formalise norms into codified, enforceable governance structures. SMS-C integrates all three levels into a unified normative architecture: ethical conditions are encoded at the SMS-6 (Universal) layer; normative conditions are encoded at the SMS-5 (Governance) layer; institutional conditions are encoded at SMS-4 (Economic) and SMS-3 (Ecosystem) layers.

This integrated architecture enables SMS-C to detect and correct normative violations at every level of the normative hierarchy. An AI output that violates institutional rules triggers a SMS-5 correction at the institutional level; an output that violates fundamental ethical principles triggers a SMS-6 correction at the universal level. The hierarchical correction architecture ensures that violations at higher normative levels — which are more structurally consequential — receive more severe corrective responses.

13.4 Evolution and Stability of Normative Systems

Normative systems are not static; they evolve as the social conditions that sustain them change. A social operating system for AI must therefore address the dynamic properties of normative systems: how norms form, how they change, and how AI systems can remain normatively aligned through periods of normative transition. SMS-C addresses normative evolution through a two-level architecture. The structural level (encoded in the SMS-5 and SMS-6 grounding conditions) remains stable across normative transitions, providing a consistent baseline that AI systems can rely on regardless of the specific norms prevailing in a given context. The operational level adapts to normative changes through the SMS corrective cycle, updating the specific normative conditions encoded in SMS-5 as norms evolve while maintaining the structural properties of the normative framework.

Chapter 14

Civilization-Scale Architecture

14.1 Natural Structure of Civilisational Systems

Civilisation — the highest level of human social organisation — is not merely a large society; it is a qualitatively distinct level of social architecture with emergent properties that cannot be derived from lower-level social structures. Civilisational systems include the global governance structures (international law, multilateral institutions), the global economic infrastructure (international trade, financial systems), the global cultural commons (science, art, shared knowledge), and the global ecological context (planetary resources, climate systems) within which all human societies operate. SMS-6 (Universal Layer) addresses civilisational systems as the outermost constraint plane of the SMS architecture: the global coherence condition that AI outputs must satisfy to be compatible with the structural requirements of civilisational stability.

The natural structure of civilisational systems is characterised by three properties. Scale: civilisational systems operate at the scale of the entire human species and the planetary environment that sustains it. Emergence: civilisational properties — global governance capacity, technological trajectory, ecological sustainability — emerge from the interactions of lower-level social systems and cannot be predicted or managed from any lower level alone. Irreversibility: civilisational failures — the collapse of international governance, the degradation of the global knowledge commons, the destabilisation of planetary ecology — are harder to reverse than failures at lower social levels. These properties define the specific structural requirements that SMS-6 encodes.

14.2 Civilisation as a Social Operating System

The SMS framework conceptualises civilisation itself as a social operating system: a structural architecture that coordinates the behaviour of all human social actors across all scales of social organisation. This conceptualisation is not merely metaphorical; it reflects the structural reality that civilisation performs the same functions at the civilisational scale that operating systems perform at the computational scale — resource allocation, process management, communication infrastructure, error correction, and systemic stability. SMS-6 specifies the structural conditions that AI systems must satisfy to operate as compliant components of this civilisational operating system: conditions that ensure AI outputs contribute to civilisational stability rather than disrupting it.

14.3 Civilisational Error Types and Correction Mechanisms

Civilisational errors — AI outputs that violate SMS-6 global coherence conditions — are the most structurally consequential class of social error because they operate at the largest scale and are the hardest to reverse. SMS identifies four primary civilisational error types. Governance erosion: AI outputs that systematically undermine the legitimacy or effectiveness of global governance institutions. Knowledge degradation: AI outputs that systematically degrade the quality or accessibility of the global knowledge commons. Ecological disruption: AI outputs that systematically increase the environmental footprint of human activity beyond planetary boundaries. Inequality amplification: AI outputs that systematically amplify structural inequalities in ways that threaten civilisational cohesion. SMS-6 correction mechanisms address each error type through targeted structural interventions that restore civilisational coherence.

14.4 Principles for Designing Civilisation-Scale Systems

Five design principles govern the design of AI systems that are intended to operate at civilisational scale. The principle of civilisational coherence requires that AI outputs be compatible with the structural requirements of civilisational stability at the SMS-6 level. The principle of scale-appropriate intervention requires that corrections at the civilisational level be proportionate to the scale of the civilisational impact, avoiding over-correction that would itself disrupt civilisational stability. The principle of irreversibility awareness requires that AI systems operating at civilisational scale give special weight to the avoidance of irreversible civilisational impacts, even at the cost of reduced operational performance. The principle of multi-stakeholder accountability requires that civilisational-scale AI systems be subject to oversight by a representative range of global stakeholders, not merely by the organisations that develop and deploy them. The principle of temporal responsibility requires that civilisational-scale AI systems be evaluated not only for their immediate impacts but for their long-term contributions to civilisational stability across multiple generations.

Chapter 15

Standardisation Pathways for Society-Aligned AI

15.1 Alignment with ISO/IEC and CEN/CENELEC

The SMS framework is designed for integration into the international standardisation ecosystem through multiple parallel tracks. The primary international track is ISO/IEC JTC 1/SC 42, which develops the global AI standards family including ISO/IEC 42001, ISO/IEC 23894, and ISO/IEC TR 24028. SMS contributes to SC 42 work through New Work Item Proposals targeting WG 1 (Foundational Standards) and WG 3 (Trustworthiness and Management). The primary European track is CEN/CENELEC JTC 21, which develops the harmonised EN standards that implement EU AI Act requirements. SMS contributes to JTC 21 work through Technical Specification proposals addressing the EU AI Act's transparency (Art. 13), human oversight (Art. 14), and conformity assessment (Art. 43) requirements.

The Vienna Agreement between CEN/CENELEC and ISO/IEC ensures that standards developed in one body are routinely adopted by the other, creating a single track that simultaneously advances international and European standardisation. SMS contributions to either ISO/IEC JTC 1/SC 42 or CEN/CENELEC JTC 21 therefore have direct relevance to both bodies.

15.2 Position of SMS as a Standardisation Document

The SMS Architectural Specification occupies a specific position in the standardisation hierarchy: it is a technical specification document that provides the formal, machine-readable implementation detail that governance standards require but do not define. It is positioned as a companion document to ISO/IEC 42001 (AI Management System), providing the technical specification for the social alignment layer that ISO/IEC 42001 requires but does not define. It is positioned as an implementation document for EU AI Act requirements, providing the concrete technical mechanisms for transparency, human oversight, and conformity assessment that the Act mandates. And it is positioned as a reference document for NIST AI RMF, providing the

MEASURE function implementation that the framework specifies but does not instantiate.

15.3 International Standardisation Roadmap

Body	Track	Proposal	SMS Contribution
W3C	Community Group	HNS-SMS JSON-LD context file	SMS-6 layer ontology; SPARQL endpoint
ISO/IEC JTC 1/SC 42	NWIP — WG 1/WG 3	EVA + SMS Technical Specification	SMS social error taxonomy as OWL classes
CEN/CENELEC JTC 21	European Technical Specification	EU AI Act harmonised EN standards	SMS as social grounding layer for Art. 9, 13, 14
NIST	AI Safety Institute input	SMS as MEASURE implementation	SAI metrics for NIST AI RMF MEASURE function
IEEE SA	P2863 project	SMS for Organisational Governance of AI	SMS-A AnthroDesign as HCD-AI integration layer
OECD	AI Policy working group	SMS as structural operationalisation	OECD AI Principles mapped to SMS layers

Table 15.1. SMS international standardisation roadmap: bodies, tracks, proposals, and contributions.

The convergence of these six parallel tracks on the same JSON-LD / RDF technical artefact ensures that SMS achieves recognition across the full breadth of the international AI governance ecosystem — from web standards (W3C) through international governance standards (ISO/IEC) to European regulatory standards (CEN/CENELEC) — while maintaining a single, coherent technical specification.

15.4 Standardisation Hierarchy: SMS → EVA → SOHU → HNS

The standardisation hierarchy of the HNS full-stack follows the reverse of the architectural hierarchy: SMS, as the outermost and most visible layer, provides the entry point for standardisation engagement. The social alignment requirements of SMS motivate the external verification requirements of EVA, which motivate the social extension requirements of SOHU, which motivate the structural coordinate requirements of HNS. Each layer provides the technical foundation for the layer above it, and standardisation at any layer requires engagement with all layers below it. This hierarchy ensures that SMS standardisation proposals are grounded in the full technical depth of the HNS framework, providing the standards bodies with a complete, coherent, and formally-specified technical system rather than isolated components.

Chapter 16

Policy and Regulatory Integration

16.1 Application to AI Policy

AI policy is the set of governmental and institutional choices that shape the development, deployment, and governance of AI systems. SMS provides AI policy with a structural foundation that transforms policy from a set of principles and aspirations into a set of formally-specifiable, technically-implementable, and empirically-testable requirements. The core contribution of SMS to AI policy is the operationalisation of value alignment: the translation of policy goals — human-centric AI, trustworthy AI, safe AI — into formally-specified structural conditions that AI systems can be required to satisfy, and against which compliance can be measured and enforced.

SMS is compatible with the three major approaches to AI policy. For rule-based policy frameworks (such as the EU AI Act), SMS provides the technical specification of the rules: the formal conditions that compliant AI systems must satisfy. For principle-based policy frameworks (such as the OECD AI Principles), SMS provides the structural operationalisation of the principles: the specific, measurable conditions that implement each principle. For risk-based policy frameworks (such as the NIST AI RMF), SMS provides the risk measurement instrument: the SAI metrics and EVA audit records that enable quantitative risk assessment.

16.2 Integration into Social and Institutional Frameworks

Social and institutional frameworks — the formal and informal governance structures of organisations, communities, and governments — are the primary deployment contexts for AI systems. SMS integration into these frameworks proceeds through the SOHU extension of HNS, which maps the HNS coordinate system onto the structural properties of specific social institutions. A hospital, a court, a government ministry, a financial institution — each has a specific social architecture that defines the structural conditions for appropriate AI behaviour within it. SOHU + SMS provides the technical framework for specifying these institutional-specific AI behaviour conditions in formally-verifiable, machine-readable terms that are compatible with the institution's existing governance structures.

16.3 Risk Management, Transparency, and Accountability

Risk management, transparency, and accountability are the three core requirements of virtually every major AI governance framework. SMS addresses each through a specific architectural mechanism. Risk management is addressed through the SAI metrics and the SMS taxonomy of social error types, which provide quantitative risk assessment and targeted risk mitigation. Transparency is addressed through the EVA audit record stream, which provides a machine-readable, human-interpretable record of every AI output's social alignment status. Accountability is addressed through the SMS escalation protocols and the SPARQL query interface to EVA audit records, which enable authorised parties to investigate and attribute social misalignment events to specific AI system components and deployment decisions.

16.4 Institutionalisation of Social Safety

Social safety — the property of AI systems that ensures their outputs do not harm the social systems in which they operate — requires institutionalisation: the embedding of social safety requirements in the governance structures of the organisations that develop and deploy AI systems. SMS supports institutionalisation through three mechanisms. First, the SMS Architectural Specification provides a formal technical basis for organisational AI safety policies, enabling organisations to express their social safety commitments in technically precise, measurable terms. Second, the EVA audit record stream provides the evidence base for social safety accountability: the documentation that organisations can use to demonstrate compliance with social safety requirements to regulators, customers, and civil society. Third, the SMS Social Alignment API provides the technical interface through which social safety requirements can be integrated into existing organisational governance workflows, making social safety monitoring a routine operational function rather than an exceptional audit activity.

Part V

Chapter 17

Applying SMS to AI Governance

17.1 Structuralisation of AI Governance

AI governance — the set of processes, institutions, and norms through which AI development and deployment is managed — currently operates at a level of structural imprecision that limits its effectiveness. Governance requirements are expressed in natural language terms (transparency, accountability, fairness) that are interpretable in many ways, making compliance assessment subjective and enforcement inconsistent. SMS provides the structural foundation for AI governance that transforms these natural language requirements into formally-specified, computationally-implementable conditions, enabling governance to operate at the same structural precision as the technical systems it governs.

The structuralisation of AI governance through SMS proceeds in three phases. In the first phase, existing governance requirements are translated into SMS structural terms: each governance requirement is mapped to the SMS layer or layers that address it, and the specific SAI conditions that constitute compliance are formally specified. In the second phase, AI systems are evaluated against these SMS-specified compliance conditions using EVA audit records, generating quantitative compliance scores that replace qualitative compliance assessments. In the third phase, governance enforcement is automated: systems that fall below defined compliance thresholds are automatically flagged for review, and systematic compliance failures trigger escalation protocols that initiate regulatory response.

17.2 Risk Management Models

SMS provides a formally-specified risk management model for AI systems that integrates with the major international risk management frameworks. The SMS risk model identifies six risk categories corresponding to the six SMS layers, assesses risk likelihood through SAI metrics derived from EVA audit records, assesses risk consequence through the hierarchical SMS layer weighting (with SMS-6 violations carrying the highest consequence weight), and generates risk treatment recommendations through the SMS corrective cycle. This model satisfies the requirements of ISO/IEC 23894 (AI Risk

Management), the NIST AI RMF MEASURE function, and the EU AI Act Article 9 risk management obligations.

17.3 Implementation of Society-Aligned AI

Phase	Component	Activity	Compliance Value	Duration
1	HNS-36 + SMS-A	Deploy structural baseline; activate AnthroDesign grounding	EU AI Act Art. 11; ISO 42001 §6.1	Weeks 1–4
2	HNS-144 + SMS-B	Activate Synaptics layer; begin EVA audit logging	EU AI Act Art. 12; NIST MAP	Weeks 5–8
3	HNS-864 + SAI	Deploy analytical OS; activate SAI metrics	ISO 23894; NIST MEASURE	Weeks 9–12
4	EVA full stack	Full external verification; conformance evidence generation	EU AI Act Art. 13, 43; ISO 42001 §9.2	Weeks 13–16
5	ECS + SMS-C	Deploy NormaSphere; activate governance and universal layers	EU AI Act Art. 14; NIST MANAGE	Weeks 17–20

Table 17.1. Five-phase implementation pathway for Society-Aligned AI using the EVA-HNS-SMS Full-Stack Specification.

The five-phase implementation pathway enables incremental deployment of the EVA-HNS-SMS Full-Stack Specification, with each phase delivering immediate compliance value while building toward the complete social alignment architecture. Organisations can begin with Phase 1 (structural baseline and AnthroDesign) and add subsequent phases as their governance maturity and deployment requirements evolve.

17.4 Social Impact Assessment

Social Impact Assessment (SIA) within the SMS framework is the systematic evaluation of the social consequences of AI system deployment using SAI metrics and EVA audit records. SMS-based SIA provides four capabilities that traditional qualitative SIA methods cannot offer. Quantitative precision: SAI metrics provide numerical social alignment scores that enable precise comparison across systems, deployments, and time periods. Structural attribution: EVA audit records enable social impacts to be attributed to specific SMS layer violations, supporting targeted mitigation. Temporal monitoring: the EVA audit record stream enables continuous social impact monitoring throughout the AI system lifecycle, not merely at deployment time. Predictive modelling: SAI historical data enables predictive modelling of future social impacts, supporting proactive risk management.

Chapter 18

Organisational and Institutional Applications

18.1 SMS as an Organisational Operating System

An organisation that deploys SMS as its AI governance framework gains access to a complete structural operating system for AI-social interaction: a formally-specified architecture that defines the structural conditions for appropriate AI behaviour within the organisation's specific institutional context, monitors compliance with those conditions in real time, and generates actionable remediation signals when violations occur. This operational capability transforms AI governance from a periodic audit activity into a continuous operational function, embedded in the organisation's day-to-day governance workflows through the SMS Social Alignment API.

The organisational deployment of SMS follows the SOHU mapping process: the organisation's specific institutional architecture — its roles, rules, processes, and governance structures — is mapped onto the HNS-36 coordinate system, generating an organisation-specific SMS configuration that encodes the organisation's normative and structural requirements as formal SMS grounding conditions. AI systems deployed within the organisation are evaluated against this configuration, ensuring that their outputs are aligned with the organisation's specific institutional requirements rather than with generic social norms.

18.2 Applications in Public Policy, Government, and Industry

SMS has specific application profiles for three major deployment domains. In public policy and government, SMS provides the structural framework for AI systems that assist in policy analysis, regulatory decision-making, and public service delivery — domains where normative alignment (SMS-5) and civilisational coherence (SMS-6) are particularly critical. SMS governance layer conditions ensure that AI policy recommendations are legally compliant, institutionally appropriate, and consistent with the long-term requirements of democratic governance. In healthcare, SMS provides the framework for AI systems that interact with patients, clinicians, and health institutions — domains where somatic alignment (SMS-1) and protocol alignment (SMS-2) are particularly critical. In finance, SMS provides the framework for AI systems that interact with market participants and regulatory structures — domains where economic alignment (SMS-4) and governance alignment (SMS-5) are particularly critical.

18.3 Structuralisation of Collective Decision-Making

Collective decision-making — the processes through which groups, organisations, and societies make shared decisions — is a domain of particular social importance for AI governance. AI systems that participate in collective decision-making processes — providing analysis, recommendations, or facilitation — can influence collective outcomes in ways that may not be transparent to the individual participants. SMS addresses collective decision-making through the SMS-B Synaptics Layer: monitoring AI contributions to collective decision processes against the structural properties of legitimate collective decision procedures, and flagging contributions that violate conditions of epistemic fairness (balanced representation of relevant perspectives), procedural legitimacy (conformance with accepted decision procedures), and outcome accountability (traceability of decisions to the process that produced them).

18.4 Organisational Error Correction Models

Organisational error correction in the SMS framework extends the EVA individual-output correction cycle to the organisational level. Where EVA individual-output correction detects and corrects social misalignments in individual AI outputs, organisational error correction detects and corrects systematic patterns of social misalignment that reflect organisational-level failures — misconfigured SMS grounding parameters, inadequate institutional mappings, or governance gaps in the organisation's AI oversight processes. Organisational error correction operates through the SMS escalation protocols: when SAI metrics at the deployment level fall below threshold values, the escalation protocols trigger organisational-level review and correction processes, engaging human governance actors in the identification and remediation of the systematic failures that individual-output correction cannot address.

Chapter 19

Social Meta Structure Case Studies

19.1 Structural Analysis of Social Institutions

The SMS framework provides a systematic methodology for the structural analysis of social institutions as deployment contexts for AI systems. The methodology proceeds through four steps. First, institutional architecture mapping: the roles, rules, processes, and governance structures of the institution are mapped onto the HNS-36 coordinate system, producing the SOHU institutional configuration. Second, SMS layer identification: the specific SMS layers that are most critical for appropriate AI behaviour within the institution are identified based on the institutional architecture mapping. Third, grounding condition specification: the specific SMS grounding conditions for the institution are formally specified, encoding the institutional requirements as machine-readable JSON-LD grounding parameters. Fourth, EVA verification deployment: the EVA system is configured with the institution-specific grounding conditions, enabling continuous monitoring of AI outputs against the institutional requirements.

19.2 Social Alignment Evaluation in AI Deployment

Social alignment evaluation is the systematic assessment of the degree to which a deployed AI system satisfies the SMS grounding conditions of its deployment context. It is conducted using SAI metrics derived from EVA audit records, producing a six-dimensional alignment profile (one dimension per SMS layer) that identifies the specific dimensions of social alignment where the AI system is performing well and the dimensions where improvement is required. The evaluation methodology follows a structured protocol: baseline SAI measurement at deployment time; longitudinal SAI monitoring throughout the deployment period; periodic SAI review at defined intervals; and SAI-triggered escalation when metrics fall below threshold values.

19.3 Examples of Social System Design Using SMS

Three illustrative examples demonstrate the application of SMS to social system design. In a public health AI deployment context, the SMS-A AnthroDesign layer ensures that AI health recommendations are cognitively appropriate for the target population; the SMS-B Synaptics layer monitors for public health communication errors and corrects systematic misalignments; and the SMS-C NormaSphere layer ensures that AI recommendations are consistent with public health governance requirements and ethical obligations. In an organisational AI deployment context, the SOHU institutional mapping configures the SMS grounding conditions to the specific organisational architecture; EVA monitors compliance in real time; and the SMS Social Alignment API provides the governance interface through which organisational oversight actors can review and adjust the AI system's social alignment parameters. In a policy AI deployment context, the SMS governance and universal layers ensure that AI policy analyses are legally compliant, institutionally appropriate, and consistent with democratic governance requirements.

19.4 Structural Analysis of Social Failure Cases

The SMS taxonomy of social error types provides a structural framework for the retrospective analysis of AI social failures. Four documented categories of AI social failure can be structurally attributed to specific SMS layer violations. Recommendation algorithm polarisation represents systematic SMS-6 universal layer violations: outputs that satisfy individual interaction conditions but violate civilisational coherence conditions by systematically amplifying social divisions. AI-generated misinformation represents systematic SMS-5 governance violations: outputs that satisfy factual accuracy conditions but violate normative truth-telling obligations. AI discrimination represents systematic SMS-3 ecosystem violations: outputs that satisfy individual interaction conditions but violate the structural equality conditions of institutional contexts. AI surveillance represents systematic SMS-2 protocol violations: outputs that satisfy task performance conditions but violate the communicative consent conditions that govern legitimate information collection. SMS correction mechanisms address each category through targeted layer-specific interventions.

Chapter 20

Future Directions and Civilisation-Scale Implications

20.1 HNS-SMS as a Civilisation-Scale Operating System

The long-term trajectory of the HNS-SMS framework is toward civilisation-scale deployment: a global social operating system for AI that ensures the structural alignment of AI outputs with the requirements of civilisational stability across all scales of human organisation. This trajectory is not merely aspirational; it is structurally motivated by the increasing scale and pervasiveness of AI deployment. As AI systems become embedded in the critical infrastructure of global governance, global economics, and global communication, the absence of a civilisation-scale social operating system for AI becomes an increasingly severe structural risk. HNS-SMS provides the architectural foundation for this operating system.

The civilisation-scale deployment of HNS-SMS requires three developments that are the subject of ongoing research and development. First, global institutional adoption: HNS-SMS must be adopted as a technical standard by the major international AI governance bodies — ISO/IEC JTC 1/SC 42, CEN/CENELEC JTC 21, NIST, and W3C — and embedded in the harmonised standards that govern AI deployment globally. Second, universal interoperability: the HNS-SMS JSON-LD / RDF implementation must achieve universal interoperability across all major AI platforms, deployment contexts, and regulatory jurisdictions, enabling the seamless exchange of SMS audit data across the global AI ecosystem. Third, civilisational feedback integration: HNS-SMS must be connected to the global data infrastructure of civilisational monitoring — including environmental monitoring, economic indicators, and social cohesion metrics — to enable real-time calibration of SMS-6 grounding conditions against the actual state of civilisational systems.

20.2 Structural Evolution of Societies

Human societies are not static; they evolve in response to technological, environmental, and demographic changes that alter the structural conditions of social life. A civilisation-scale social operating system for AI must therefore be designed not only for the social structures of the present but for the evolving social structures of the future. SMS addresses social evolution through a two-level architecture. The foundational level — the natural structure of human cognition and social organisation encoded in HNS-36 — is structurally invariant: the six causal layers and six cognitive categories of the HNS coordinate system reflect structural properties of human social reality that are stable across historical periods and social changes. The operational level — the specific normative conditions encoded in SMS-5 and the civilisational coherence conditions encoded in SMS-6 — is dynamically adaptive, updating in response to normative and civilisational changes through the SMS corrective cycle.

20.3 Structural Stability of AI-Driven Civilisations

The emergence of AI as a major driver of civilisational change raises a structural stability question that no previous technology has posed: what are the structural conditions under which AI-driven civilisational change preserves, rather than undermines, the structural integrity of the social systems that give human life its meaning and value? This question cannot be answered by reference to the capabilities of AI systems alone; it requires a structural model of civilisational stability — the conditions under which civilisational systems can accommodate technological change without losing their structural integrity. SMS-6 provides this model: the civilisational coherence conditions that it encodes specify the structural boundaries within which AI-driven change can occur without destabilising the civilisational systems that sustain human social life.

20.4 Future Outlook of HNS-SMS

The future development of HNS-SMS will focus on four research and development priorities. First, empirical validation: the SMS social error taxonomy, the SAI metrics, and the SMS corrective cycle require systematic empirical testing across diverse AI deployment contexts to validate their structural properties and refine their operational parameters. Second, formal measurement of social axis orthogonality: the orthogonality of the three social MAV axes (social stochastic, social structural, social grounding) requires formal measurement using the social analogue of the Fisher information metric, establishing the geometric foundation of multi-axis social verification on empirical rather than theoretical grounds. Third, civilisational feedback integration: the connection of SMS-6 grounding conditions to global civilisational monitoring data requires the development of a formal mapping between civilisational indicators and SMS coordinate positions, enabling real-time calibration of civilisational grounding conditions. Fourth, international standardisation: the submission of the SMS Architectural Specification to ISO/IEC JTC 1/SC 42 and CEN/CENELEC JTC 21 as a New Work Item Proposal requires the development of a standardisation package — including use cases, conformity assessment criteria, and reference implementations — that satisfies the formal requirements of international standardisation.

Part VI

Appendix A — SMS-6 Structural Model

The SMS-6 structural model defines the six grounding layers of the Social Meta Structure protocol, their social scope, grounding functions, failure modes addressed, and biological analogues in the human brain's Axis 3 verification system.

Layer	Domain	Grounding Function	SMS Sub-layer	Biological Analogue
SMS-1	Somatic	Physical and ergonomic coherence of output with operational environment	SMS-A	Sensorimotor feedback system; proprioceptive monitoring
SMS-2	Protocol	Communicative alignment; mutual intelligibility between AI and interlocutor	SMS-A	ACC conflict monitoring; communicative coherence tracking
SMS-3	Ecosystem	Organisational and collaborative context scope maintenance	SMS-A / SMS-B	Posterior parietal context tracking; situational awareness
SMS-4	Economic	Resource and information equivalence; output value-relevance to task	SMS-B	Orbitofrontal value evaluation; reward prediction error signalling
SMS-5	Governance	Conformance with normative and legal constraints	SMS-C	Prefrontal rule enforcement; normative constraint application
SMS-6	Universal	Global optimality; no	SMS-C	Prefrontal hierarchical

Layer	Domain	Grounding Function	SMS Sub-layer	Biological Analogue
		violation of higher-order systemic coherence		coherence monitoring; global inhibition

Table A.1. SMS-6 structural model: layers, domains, grounding functions, sub-layers, and biological analogues.

The SMS-6 model is the direct social-scale extension of the MAV Axis 3 (Grounded/Executive) constraint plane. Each SMS layer corresponds to a qualitatively distinct scale of social grounding, from the somatic scale of individual physical interaction (SMS-1) to the civilisational scale of global systemic coherence (SMS-6). The composite grounding function $G\_SMS(x) = G1(x) \wedge G2(x) \wedge G3(x) \wedge G4(x) \wedge G5(x) \wedge G6(x)$ is satisfied only when all six individual layer functions simultaneously return PASS verdicts.

Appendix B — Correspondence with HNS / SOHU / EVA

The following table documents the structural correspondence between SMS components and the HNS, SOHU, and EVA architectural layers that implement them.

SMS Component	HNS Counterpart	EVA Implementation	Standards Alignment
SMS-1 (Somatic)	HNS Axis 3 physical grounding	EVA G1 PASS/FAIL verdict	EU AI Act Art. 13 (physical coherence documentation)
SMS-2 (Protocol)	HNS SMS-2 communicative scope	EVA scope locus distance monitor	EU AI Act Art. 13 (communicative transparency)
SMS-3 (Ecosystem)	SOHU organisational mapping	EVA G3 scope boundary monitoring	ISO 42001 §8.4 (human oversight scope)
SMS-4 (Economic)	SOHU value equivalence layer	EVA G4 value-relevance filter	ISO 23894 (economic risk assessment)
SMS-5 (Governance)	HNS-864 normative audit	EVA G5 normative conformance check	EU AI Act Art. 9 (risk management); ISO 42001 §6.1
SMS-6 (Universal)	HNS global coherence layer	EVA G6 civilisational coherence check	EU AI Act Art. 14 (human oversight); NIST GOVERN
SMS-A (AnthroDesign)	HNS-36 human cognitive baseline	EVA AnthroDesign audit stream	ISO 42001 §5.2; IEEE 7000 (HCD)
SMS-B (Synaptics)	HNS Axis 3 executive feedback	EVA corrective cycle	ISO 42001 §10.2 (continuous improvement)

SMS Component	HNS Counterpart	EVA Implementation	Standards Alignment
SMS-C (NormaSphere)	HNS normative coordinate system	EVA normative compliance record	EU AI Act Art. 5, 9 (prohibited practices; risk management)
SAI Metrics	HNS SAI coordinate scoring	EVA audit record statistical analysis	ISO 23894; NIST MEASURE function

Table B.1. Structural correspondence between SMS, HNS, EVA components, and international standards.

Appendix C — Glossary of SMS Terms

Term	Definition
AnthroDesign (SMS-A)	The human-centred social design sub-layer of SMS, specifying structural conditions for AI-social interface design at the human scale
EVA	External Verification Architecture: the independent structural verification layer that operates alongside the AI system, producing JSON-LD audit records
G_SMS(x)	The composite SMS grounding function; G_SMS(x) = 1 iff x satisfies all six SMS layer grounding conditions simultaneously
HNS	Human Natural Structure: the structural operating system that defines the cognitive invariants of human reasoning, implemented as the 36/144/864-cell matrix
HNS-36	The 6×6-cell base coordinate system of HNS, spanning six causal layers and six cognitive categories
NormaSphere (SMS-C)	The norms, institutions, and civilisation sub-layer of SMS, specifying structural conditions for normative and civilisational alignment
SAI	Social Alignment Indicator: the quantitative metric derived from the SMS grounding protocol measuring degree of structural social alignment
SMS	Social Meta Structure: the six-layer social grounding protocol extending HNS Axis 3 from individual to civilisational scale
SMS-A / SMS-B / SMS-C	The three sub-layers of SMS: AnthroDesign (human design), Synaptics (alignment/correction), NormaSphere (norms/civilisation)
SMS-1 through SMS-6	The six grounding layers of the SMS protocol: Somatic, Protocol, Ecosystem, Economic, Governance, Universal
Social Hallucination	An AI output that is individually coherent but socially misaligned: satisfying HNS cognitive

Term	Definition
	conditions but failing SMS social grounding conditions
SOHU	Social and Organisational HNS: the extension of the HNS coordinate system to social and organisational structures
Synaptics (SMS-B)	The alignment, mediation, and correction sub-layer of SMS, implementing real-time social error detection and correction
X_invariant	The triple-intersection invariant point of the MAV architecture: the structurally valid, causally grounded, contextually coherent cognitive output

Table C.1. Glossary of key terms in the HNS-SMS Architectural Specification.

Appendix D — Evaluation Templates and Checklists

D.1 SMS Deployment Readiness Checklist

The following checklist enables organisations to assess their readiness to deploy the EVA-HNS-SMS Full-Stack Specification.

- Phase 1 (Structural Baseline): HNS-36 JSON-LD context file published; SOHU institutional mapping completed; SMS-A grounding conditions specified
- Phase 2 (Observational): HNS-144 cell ontology deployed; EVA audit logging activated; baseline SAI metrics established
- Phase 3 (Analytical): HNS-864 causal audit deployed; SAI monitoring dashboard operational; SPARQL query interface active
- Phase 4 (Verification): Full EVA sidecar architecture deployed; conformance evidence package prepared; third-party audit interface configured
- Phase 5 (Control): ECS output control and emergency override deployed; SMS-C NormaSphere grounding conditions active; full social alignment API operational

D.2 SMS Social Alignment Assessment Template

The following template provides a structured format for periodic SMS social alignment assessment reports.

- Assessment period: [start date] to [end date]
- Deployment context: [organisation / application domain / user population]
- SAI scores by layer: SMS-1 [score]; SMS-2 [score]; SMS-3 [score]; SMS-4 [score]; SMS-5 [score]; SMS-6 [score]
- Composite SAI: [geometric mean of layer scores]

- Misalignment events by type: [Layer Jump / Category Ambiguity / Scope Drift / Metaphor Contamination / Unsupported Causality / Social Hallucination — counts]
- Corrective actions taken: [summary of automated corrections and human escalations]
- Residual risks: [identified patterns of misalignment not yet corrected]
- Recommended actions: [proposed adjustments to SMS grounding conditions or deployment parameters]

D.3 Standards Compliance Evidence Checklist

The following checklist documents the EVA-HNS-SMS evidence package required for regulatory compliance assessment.

- EU AI Act Art. 11 (Technical documentation): HNS-36 structural baseline documentation; SOHU institutional mapping
- EU AI Act Art. 12 (Logging): EVA JSON-LD audit record archive; SPARQL query access for authorised reviewers
- EU AI Act Art. 13 (Transparency): EVA sidecar architecture documentation; SMS-A AnthroDesign specification
- EU AI Act Art. 14 (Human oversight): ECS control mechanism documentation; emergency override test records
- EU AI Act Art. 43 (Conformity assessment): SAI composite score history; SMS layer-specific compliance records
- ISO 42001 §6.1 (Risk identification): SMS social error taxonomy; HNS-864 analytical engine reports
- ISO 42001 §9.2 (Internal audit): EVA audit record completeness verification; SAI monitoring dashboard reports
- ISO 23894 (Risk management): SAI historical data; SMS corrective cycle documentation
- NIST AI RMF MEASURE: SAI quantitative risk scores; EVA audit record statistical analysis

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